
PRACTICAL - 6

Implementation of Matrix Chain Multiplication using Dynamic Programming

Aim of the Practical:

To find the parenthesization with minimum scalar multiplications using dynamic programming and derive its complexity. The program also prints the matrix-cost table, split-position table, every evaluated split, and complete bracket reconstruction.

Recurrence:

$$m[i,j] = 0, \text{ if } i = j$$

$$m[i,j] = \min\{i \leq k < j\} [m[i,k] + m[k+1,j] + p[i-1]p[k]p[j]]$$

Program:

```
#include <climits>
#include <iomanip>
#include <iostream>
#include <sstream>
#include <string>
#include <vector>

using namespace std;

string parenthesize(int i, int j, const vector<vector<int>>& split) {
    if (i == j) return "A" + to_string(i);
    int k = split[i][j];
    return "(" + parenthesize(i, k, split) + " x "
        + parenthesize(k + 1, j, split) + ")";
}

void printCostTable(const vector<vector<long long>>& cost, int n) {
    cout << "\nM TABLE (minimum scalar multiplications)\n";
    cout << left << setw(8) << "i\\j";
    for (int j = 1; j <= n; ++j) cout << setw(10) << j;
    cout << '\n';

    for (int i = 1; i <= n; ++i) {
        cout << left << setw(8) << i;
        for (int j = 1; j <= n; ++j) {
            if (j < i) cout << setw(10) << "-";
            else cout << setw(10) << cost[i][j];
        }
        cout << '\n';
    }
}

void printSplitTable(const vector<vector<int>>& split, int n) {
    cout << "\nK TABLE (best split position)\n";
    cout << left << setw(8) << "i\\j";
    for (int j = 1; j <= n; ++j) cout << setw(10) << j;
    cout << '\n';

    for (int i = 1; i <= n; ++i) {
        cout << left << setw(8) << i;
        for (int j = 1; j <= n; ++j) {
            if (j <= i) cout << setw(10) << "-";
            else cout << setw(10) << split[i][j];
        }
        cout << '\n';
    }
}

int main() {
    const vector<int> p = {4, 10, 3, 8};
    const int n = static_cast<int>(p.size()) - 1;

    vector<vector<long long>> cost(n + 1, vector<long long>(n + 1, 0));
    vector<vector<int>> split(n + 1, vector<int>(n + 1, 0));

    cout << "MATRIX CHAIN MULTIPLICATION USING DYNAMIC PROGRAMMING\n";
```

```

cout << "-----\n";
cout << "Dimension array p = {4, 10, 3, 8}\n\n";
for (int i = 1; i <= n; ++i) {
    cout << "A" << i << " = " << p[i - 1] << " x " << p[i] << '\n';
}

cout << "\nSTEP-BY-STEP DP CALCULATION\n";
for (int length = 2; length <= n; ++length) {
    cout << "\nChain length = " << length << '\n';
    for (int i = 1; i <= n - length + 1; ++i) {
        int j = i + length - 1;
        cost[i][j] = LLONG_MAX;
        cout << "\nFinding m[" << i << "][" << j << "] for A"
            << i << "...A" << j << ":\n";

        for (int k = i; k < j; ++k) {
            long long leftCost = cost[i][k];
            long long rightCost = cost[k + 1][j];
            long long multiplyCost = 1LL * p[i - 1] * p[k] * p[j];
            long long total = leftCost + rightCost + multiplyCost;

            cout << "    k = " << k << ": m[" << i << "][" << k << "] + m["
                << k + 1 << "][" << j << "] + p[" << i - 1 << "]*p["
                << k << "]*p[" << j << "]\n";
            cout << "    = " << leftCost << " + " << rightCost << " + "
                << p[i - 1] << "*" << p[k] << "*" << p[j] << '\n';
            cout << "    = " << leftCost << " + " << rightCost << " + "
                << multiplyCost << " = " << total << '\n';

            if (total < cost[i][j]) {
                cost[i][j] = total;
                split[i][j] = k;
                cout << "        -> New minimum; set m[" << i << "][" << j
                    << "] = " << total << " and k[" << i << "][" << j
                    << "] = " << k << '\n';
            } else {
                cout << "        -> Not smaller than current minimum "
                    << cost[i][j] << '\n';
            }
        }
    }
}

printCostTable(cost, n);
printSplitTable(split, n);

string optimal = parenthesize(1, n, split);
cout << "\nHOW THE BRACKETS ARE FORMED\n";
cout << "k[1][3] = " << split[1][3]
    << ", so split A1...A3 after A" << split[1][3] << ".\n";
cout << "Left part A1...A" << split[1][3]
    << " becomes " << parenthesize(1, split[1][3], split) << ".\n";
cout << "Right part A" << split[1][3] + 1 << "...A3 becomes "
    << parenthesize(split[1][3] + 1, 3, split) << ".\n";
cout << "Combine both parts: " << optimal << '\n';

cout << "\nOPTIMAL MULTIPLICATION ORDER\n";
cout << "Step 1: (A1 x A2) = (4 x 10) x (10 x 3)\n";
cout << "    Result dimension = 4 x 3\n";
cout << "    Scalar multiplications = 4*10*3 = 120\n";
cout << "Step 2: ((A1 x A2) x A3) = (4 x 3) x (3 x 8)\n";
cout << "    Result dimension = 4 x 8\n";
cout << "    Scalar multiplications = 4*3*8 = 96\n";
cout << "Total = 120 + 96 = " << cost[1][n] << '\n';

cout << "\nFINAL ANSWER\n";
cout << "Optimal parenthesization: " << optimal << '\n';
cout << "Minimum scalar multiplications: " << cost[1][n] << '\n';
cout << "Time complexity: Theta(n^3)\n";
cout << "Space complexity: Theta(n^2)\n";
return 0;
}

```

Output:

MATRIX CHAIN MULTIPLICATION USING DYNAMIC PROGRAMMING

Dimension array p = {4, 10, 3, 8}

A1 = 4 x 10
A2 = 10 x 3
A3 = 3 x 8

STEP-BY-STEP DP CALCULATION

Chain length = 2

Finding m[1][2] for A1...A2:

k = 1: m[1][1] + m[2][2] + p[0]*p[1]*p[2]
= 0 + 0 + 4*10*3

$= 0 + 0 + 120 = 120$
 -> New minimum; set $m[1][2] = 120$ and $k[1][2] = 1$

Finding $m[2][3]$ for $A2...A3$:

$k = 2: m[2][2] + m[3][3] + p[1]*p[2]*p[3]$
 $= 0 + 0 + 10*3*8$
 $= 0 + 0 + 240 = 240$
 -> New minimum; set $m[2][3] = 240$ and $k[2][3] = 2$

Chain length = 3

Finding $m[1][3]$ for $A1...A3$:

$k = 1: m[1][1] + m[2][3] + p[0]*p[1]*p[3]$
 $= 0 + 240 + 4*10*8$
 $= 0 + 240 + 320 = 560$
 -> New minimum; set $m[1][3] = 560$ and $k[1][3] = 1$
 $k = 2: m[1][2] + m[3][3] + p[0]*p[2]*p[3]$
 $= 120 + 0 + 4*3*8$
 $= 120 + 0 + 96 = 216$
 -> New minimum; set $m[1][3] = 216$ and $k[1][3] = 2$

M TABLE (minimum scalar multiplications)

i\j	1	2	3
1	0	120	216
2	-	0	240
3	-	-	0

K TABLE (best split position)

i\j	1	2	3
1	-	1	2
2	-	-	2
3	-	-	-

HOW THE BRACKETS ARE FORMED

$k[1][3] = 2$, so split $A1...A3$ after $A2$.

Left part $A1...A2$ becomes $(A1 \times A2)$.

Right part $A3...A3$ becomes $A3$.

Combine both parts: $((A1 \times A2) \times A3)$

OPTIMAL MULTIPLICATION ORDER

Step 1: $(A1 \times A2) = (4 \times 10) \times (10 \times 3)$
 Result dimension = 4×3
 Scalar multiplications = $4*10*3 = 120$
 Step 2: $((A1 \times A2) \times A3) = (4 \times 3) \times (3 \times 8)$
 Result dimension = 4×8
 Scalar multiplications = $4*3*8 = 96$
 Total = $120 + 96 = 216$

FINAL ANSWER

Optimal parenthesization: $((A1 \times A2) \times A3)$

Minimum scalar multiplications: 216

Time complexity: $\Theta(n^3)$

Space complexity: $\Theta(n^2)$

Matrix Cost Table (m):

$m[i,j]$	$j = 1$	$j = 2$	$j = 3$
$i = 1$	0	120	216
$i = 2$	-	0	240
$i = 3$	-	-	0

Split Position Table (k):

$k[i,j]$	$j = 1$	$j = 2$	$j = 3$
$i = 1$	-	1	2
$i = 2$	-	-	2
$i = 3$	-	-	-

Step-by-Step Cost Calculation for the Sample:

- $A1 = 4 \times 10$, $A2 = 10 \times 3$, $A3 = 3 \times 8$.

- $m[1,2]$: $k = 1$ gives $0 + 0 + (4 \times 10 \times 3) = 120$.

- $m[2,3]$: $k = 2$ gives $0 + 0 + (10 \times 3 \times 8) = 240$.

- $m[1,3], k = 1: 0 + 240 + (4 \times 10 \times 8) = 560.$
- $m[1,3], k = 2: 120 + 0 + (4 \times 3 \times 8) = 216$ (minimum).

How the Brackets Are Formed:

The value $k[1,3] = 2$ places the outer split after A_2 . The left subchain $A_1 \dots A_2$ uses $k[1,2] = 1$, so it becomes $(A_1 \times A_2)$. The right subchain contains only A_3 . Combining them places the brackets as $((A_1 \times A_2) \times A_3)$.

```

A1...A3
  split after A2
    /         \
  (A1 x A2)   A3
    \         /
  ((A1 x A2) x A3)

```

Order of Matrix Multiplication:

Step 1: Multiply $A_1(4 \times 10)$ by $A_2(10 \times 3)$. The result is 4×3 and costs $4 \times 10 \times 3 = 120$ scalar multiplications.

Step 2: Multiply the result (4×3) by $A_3(3 \times 8)$. The final result is 4×8 and costs $4 \times 3 \times 8 = 96$ scalar multiplications.

Total cost = $120 + 96 = 216$ scalar multiplications.

Time Analysis:

Derivation of Time Complexity:

Let n be the number of matrices. The outer loop chooses chain length, the next loop chooses the starting index, and the inner loop checks every possible split point.

$$T(n) = \sum(\text{len}) \sum(i) \sum(k) O(1)$$

$$T(n) = O(n \times n \times n) = O(n^3)$$

The cost and split tables each have $(n + 1)^2$ entries, so $S(n) = \Theta(n^2)$.

Final Time Analysis:

- Time Complexity: $\Theta(n^3)$.
- Space Complexity: $\Theta(n^2)$.

Result:

Matrix Chain Multiplication was implemented successfully. The cost table, split table, evaluated costs, and bracket construction confirm that the optimal parenthesization is $((A_1 \times A_2) \times A_3)$, with a minimum cost of 216 scalar multiplications.